

Proposal for:

A Tale of Two Anomalies: The Implication of Investor
Attention for Price and Earnings Momentum

by

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Abstract

In this project, we examine the profitability of price and earnings momentum strategies. We find that price momentum profits are higher among high volume stocks and in up markets, while earnings momentum profits are higher among low volume stocks and in down markets. In the long run, price momentum profits are reversed, while earnings momentum profits are not. The dichotomy between price and earnings momentum is more pronounced when we orthogonalize one with respect to the other. To the extent that trading volume increases with investor attention and that investors tend to pay more attention to stocks in up markets, our results suggest a dual role for investor attention: while price underreaction to earnings news declines with investor attention, price continuation caused by investors' overreaction rises with attention.

1 Motivation and Research Objectives

Attention is a scarce cognitive resource (Kahneman, 1973). A large body of psychological research shows that there exists a limit to the central cognitive-processing capacity of the human brain. The inevitability of limited attention in relation to the vast amount of information available makes attention an important factor in agents' learning and decision-making processes. There is now a growing literature analyzing the economic consequences of limited investor attention. Hirshleifer and Teoh (2003) analyze firms' accounting disclosure policy in the presence of inattentive investors. Barber and Odean (2005) study how salient events can capture investors' attention and affect their stock buying and selling decisions. Peng (2005) and Peng and Xiong (2006) analyze the effects of limited attention on investors' learning behavior and equilibrium price dynamics.

In this project, we examine how attention affects asset price dynamics through investors' under- and overreaction to information – two basic mechanisms developed in the finance literature to explain a large body of empirical anomalies in asset return predictability. Specifically, we analyze the role of investor attention in two widely documented anomalies – price momentum and earnings momentum (also known as post-earnings announcement drift).

Starting with Jegadeesh and Titman (1993), many studies, e.g., Rouwenhorst (1998) and Griffin, Ji, and Martin (2003), and Grundy and Martin (2001), find that a trading strategy based on buying recent winners over the past 3-12 months and simultaneously shorting recent losers over the same period can provide a significant trading profit in both U.S. and international stock markets. This phenomenon is often referred to as price momentum. Several other studies, e.g., Ball and Brown (1968), Bernard and Thomas (1989), and Chan, Jegadeesh and Lakonishok (1996), find that for 60 days after earnings announcements, returns of NYSE/AMEX stocks continue to drift up for “good news” firms and down for “bad news” firms. This phenomenon is often referred to as post-earnings announcement drift or earnings momentum. The empirical literature also finds that risk premium cannot explain either price or earnings momentum.

Barberis, Shleifer and Vishny (1998) and Hong and Stein (1999) model investors' underreaction to information. In these models, investors underreact to news about firm funda-

mentals, resulting in insufficient initial price reaction to the news. Meanwhile, the process of news gradually being incorporated into prices in the following periods, generates both price and earnings momentum. Along a different direction, De Long et al (1990) and Daniel, Hirshleifer and Subrahmanyam (1998) focus on investors' overreactions. In these models, behavioral biases, such as extrapolative expectation and self attribution, cause investors, after observing recent rises, to further push up stock prices, also leading to both price and earnings momentum.

Thus under- and overreaction-based mechanisms give similar predictions for both price and earnings momentum. This similarity in prediction of the two mechanisms makes it hard to differentiate their relative contribution in explaining these anomalies. We sharpen the implications by exploring the dual role played by investor attention in under- and overreaction to information. On the one hand, inadequate attention directly causes ignorance of useful information and therefore stock price underreaction; on the other hand, attention can interact with investors' behavioral biases, such as extrapolative expectations and overconfidence, to generate price overreaction. This dual role of attention allows us to develop contrasting empirical hypotheses in both cross-section and time-series for under- and overreaction-based mechanisms. We test these hypotheses and provides quantitative assessment of the contribution of under- and overreaction-based mechanisms to price and earnings momentum.

2 Empirical Hypotheses

Limited attention imposes a constraint on the amount of information that investors can process and to which they can react to. Consequently, attention-limited investors may ignore useful public information. Hirshleifer and Teoh (2005), Peng (2005), and Peng and Xiong (2006) provide models to analyze this effect. These models suggest that, when investors' attention to a firm is inadequate, they may ignore its earnings announcement, resulting in stock price underreaction to the earnings news. After the announcement, the price continues to drift in the same direction as the information eventually gets incorporated. Thus, investor inattention gives rise to earnings momentum. Furthermore, the magnitude of the earnings momentum decreases investor attention.

Limited attention is not itself a behavioral bias, but it could interact with biases and

thus influence how investors react to the processed information. In particular, attention is a necessary ingredient in investors' overreaction to information (Peng and Xiong, 2006). Thus, investor attention affects the overreaction-driven price and earnings momentum, suggested by De Long, et al (1990) and Daniel, Hirshleifer and Subrahmanyam (1998). If investors do not pay attention to a stock, they can neither over-extrapolate the stock's past returns, nor overreact to their private information. Consequently, there will be no overreaction-driven price momentum. Conversely, when investors pay more attention to a stock, these biases can generate stronger price momentum.

In summary, investor attention plays a dual role in stock price under- and overreactions: while inadequate attention directly generates price underreaction to earnings news and therefore earnings momentum, the interaction between attention and investors' learning biases (extrapolative expectations, overconfidence), leads to overreaction-driven price momentum. In a cross-section of stocks, we expect those that receive *more* investor attention to display *stronger* overreaction-driven price momentum, but *weaker* earnings momentum.

We use trading volume as a proxy of investor attention in our cross-sectional analysis. On the one hand, investors cannot actively trade a stock if they do not pay attention to it; on the other, when investors do pay attention, behavioral biases such as overconfidence can give rise to heterogeneous opinions among investors about asset fundamentals, thus generating more trading (Odean, 1998 and Scheinkman and Xiong, 2003).¹ Using trading volume as a proxy for investor attention, we obtain the following testable hypothesis:

Hypothesis I. In a cross-section of stocks, those with higher trading volume tend to display *stronger* price momentum, but *weaker* earnings momentum.

We further expect that overreaction-driven price momentum would be reversed in the long run, as price overreaction is eventually corrected. In contrast, if earnings momentum is

¹There is evidence supporting the link between trading volume and investor attention. Lo and Wang (2000) show that trading volume tends to be higher among large stocks, consistent with the fact that large stocks tend to attract more investor attention. Chordia and Swaminathan (2000) find that, controlling for firm size, returns of high volume portfolios lead returns of low volume portfolios, and this finding cannot be explained by nonsynchronous trading. They show that low volume stocks respond slower to public information in market returns, suggesting that low volume stocks tend to be those that receive less attention and that trading volume is a better measure of investor attention than firm size. Gervais, Kaniel and Mingelgrin (2001) and Barber and Odean (2005) also use volume as an attention proxy.

caused by investor inattention, the price pattern will not be reversed in the long run.

The attention that investors allocate to stocks not only varies in cross-section, but also across time. Karlsson, Loewenstein and Seppi (2005) analyze account activity in three Scandinavian data sets: the daily number of investor account look-ups at a large Norwegian financial service company, the daily number of online logins of a major Swedish bank, and the daily number of pension account look-ups by investors of the Swedish Pension Authority. They find that investors are more likely to look up their portfolio in up markets than in down markets. This “ostrich effect” suggests that investors pay more attention to stocks in rising markets, but “put their heads in the sand” in flat or falling markets.

The increased attention in up markets can cause investors to overreact more to their private information and to past returns, generating a more pronounced pattern of overreaction-driven price momentum. The increased attention also means that firms’ earnings announcements are less likely to be ignored by investors, causing weaker earnings momentum in up markets. We summarize the time-series properties of price and earnings momentum in the following hypothesis:

Hypothesis II. Price momentum is *stronger* in *up* markets than in down markets, while earnings momentum is *weaker* in up markets than in down markets.

3 Research Methodology

Our data include all NYSE/AMEX listed securities from October, 1971 to December, 2000. We measure trading volume using the average monthly turnover over the one year period prior to the portfolio formation month. The monthly turnover is the number of shares traded during a month divided by the number of shares outstanding at the end of the month. We acquire quarterly earnings data from COMPUSTAT. To avoid using stale earnings, a firm has to have the most recent earnings announcement within four months prior to the portfolio formation month. Following Chan, Jegadeesh and Lakonishok (1996), we measure earnings surprise by the standardized unexpected earnings (SUE). The SUE for stock i in month t is defined as

$$SUE_{i,t} = \frac{e_{i,t} - e_{i,t-4}}{\sigma_{i,t}}$$

where $e_{i,t}$ is quarterly earnings per share most recently announced as of month t for stock i , $e_{i,t-4}$ is earnings per share four quarters ago, and $\sigma_{i,t}$ is the standard deviation of unexpected earnings, $e_{i,t} - e_{i,t-4}$, over the last eight quarters. We employ the market excess returns, the SMB factor (the factor mimicking returns for the size effect), and the HML factor (the factor mimicking returns for the BE/ME effect) in some of our tests.

To examine the relationship between price momentum and volume, we form stock portfolios double-sorted by turnover and past returns. At the beginning of each month, we sort all NYSE/AMEX stocks in our sample into quintiles based on their average monthly turnover over the previous year. Within each turnover quintile, we further sort stocks into quintiles based on return over the past twelve months (skipping the most recent month to avoid market microstructure effects).² We then compute equal-weighted returns of these portfolios over the following month. The return spread between the winner and loser portfolios (past return quintiles 5 and 1 within each turnover quintile) constitutes the profit from the price momentum strategy. Part of the price momentum profit could be caused by investors' underreaction to the earnings surprises. This is because past winners might have recently had positive earnings surprises, while past losers might have recently had negative ones. To control for this possibility, we run cross-sectional regressions of the past 12-month stock returns onto the most recent unexpected earnings (SUE) and use the residual returns as the sorting variable to form price momentum portfolios.

To analyze the relationship between earnings momentum and volume, we form portfolios double-sorted by turnover and the unexpected earnings (SUE). Each month, we first sort stocks into quintiles based on their turnover. Within each turnover quintile, we further group stocks into quintiles based on their most recent SUE. An earnings momentum strategy is to buy stocks in the highest SUE quintile and simultaneously short stocks in the lowest SUE quintile. The profit of this strategy is the return spread between the highest and the lowest SUE quintiles. Like price momentum profits, the observed earnings momentum profits can be partially driven by price continuation related to investors' overreaction to past returns. To control for this component, we run cross-sectional regressions of SUE onto the past one-year

²Earlier studies, e.g., Jegadeesh and Titman (1993), find that alternative strategies with portfolio formation periods ranging from 1 to 4 quarters and holding periods from 1 to 12 months provide similar trading profits.

return, and use the regression residual to form earnings momentum portfolios.

We analyze whether the price and earnings momentum profits are related to investors' under- or overreactions by examining the long run performance of these strategies. We compute profits for three additional holding periods, months 1-6, 1-12, and 13-60 after the portfolio formation month. To control for risk in the momentum profits, we regress both price and earnings momentum profits against the excess return of the Fama-French three risk factors. We also analyze the characteristics-adjusted returns on the turnover and past-return double-sorted portfolios, as well as the momentum profits computed from the characteristics-adjusted returns.

To analyze the time-series properties of price and earnings momentum profits across up and down markets, we define market state based on the recent performance of the value-weighted CRSP index (including dividends) over the most recent 36 months or 24 months. We define a month as "UP" if the recent CRSP index return is nonnegative, and as "DOWN" otherwise. We compute the price and earnings momentum profits and compare the average profits across UP and DOWN months. We also regress the price and earnings momentum profits against the Fama-French three risk factor to control for risk premium.

4 Preliminary Results

We find that price momentum profits, in both raw and risk-adjusted returns, monotonically increase with volume, and are significant across all quintiles. The difference between the risk adjusted price momentum profits in the highest and lowest volume quintiles is both statistically and economically significant with a value of 66 basis points per month. Controlling for unexpected earnings causes a slight drop in the price momentum profits, but does not change the monotonically increasing relationship between price momentum profits and volume. We also find some evidence of reversals – the long-run returns of the price momentum portfolios for months 13-60 after the portfolio formation are negative across all volume quintiles and in both raw and earnings news adjusted measures. Overall, these results suggest that there is a significant overreaction-driven component in price momentum profits and this component monotonically increases with trading volume, consistent with our attention-based hypothesis.

We also find that the earnings momentum profits are strongest in the lowest volume

quintile and weakest in the highest volume quintile. The difference becomes even larger after controlling for prior stock returns. The risk adjusted profit in the lowest volume quintile is 77 basis points per month higher than that in the highest quintile, and this difference is significant with a p-value of 0.02%. Controlling for prior returns also causes a sizable drop in earnings momentum profits, suggesting that the uncontrolled earnings momentum profits is partially related to price momentum. Furthermore, the long-run returns of the earnings momentum portfolios in years 2-5 show no sign of reversals. These results support our hypothesis that investors' underreaction to earnings news contributes to earnings momentum and the degree of underreaction is decreasing with investor attention.

In our time-series analysis, we find that price momentum profits barely exist in DOWN months, but are significantly positive in UP months. The difference, more than 120 basis points per month, is statistically significant. Quite the reverse, the monthly earnings momentum profits, after controlling for prior stock returns, are significantly higher in DOWN months than in UP months with a difference of 72 basis points. These opposing patterns in price and earnings momentum profits across UP and DOWN months directly support our attention-based hypothesis.

5 Practical Value

The profitability of price and earnings momentum strategies in different time periods and in different stock groups should be of great interest to investment practitioners. Our analysis shows that the profitability of these strategies varies dramatically! The profit differential of price momentum is as large as 66 basis points per month across volume quintiles, and 120 basis points per month across up and down markets. These patterns would allow practitioners to sharpen their momentum strategies.

What's more, these patterns in price and earnings momentum come from a natural and believable story (both from an academic and practitioner point of view). Most practitioners and academics acknowledge that investors only possess limited mental attention. A direct consequence is that investors can only pay attention to one set of assets and ignore others. Our study exactly builds on this simple notion, and shows that it leads to sharp implications for two popular investment strategies. This compelling story provides a simple rationale that

almost any practitioner can understand. This, we think, would provide a nice marketing edge associated with attention-based momentum strategies.

Finally, our attention-based story can also be applied to understand and sharpen other types of trading strategies that are related to investors' under and overreactions. For example, academic studies have documented profitable strategies by sorting on firms' accounting accruals or intangible assets. Many practitioners are concerned about whether these strategies are generated by data mining. Our attention-based story can be extended to provide not only an explanation of these phenomena, but also sharp predictions about the cross-sectional and time-series properties of the profitability of these strategies.

6 Future Plans and Timetable

Our proposed time-series measure of investor attention, based on the up and down state of the market index over the past 24 or 36 months, is fairly simple. We plan to explore other measures based on the economic recession periods defined by NBER, and on output rate. We also plan to further analyze volume as a cross-sectional measure of investor attention. In addition, earlier studies have shown that across stocks sorted by firm size and analyst coverage, price and earnings momentum profits vary and are positively correlated. It is an interesting exercise to examine results across different firm characteristics, such as size, analyst coverage and volume.

We also plan to extend our study to examine the contribution of different investor clienteles, i.e., institutions' or retail investors', to our results. To address this question, we can use institutional ownership to proxy for the degree that institutions are the marginal investor in a firm and analyze the interaction between attention effects and institutional ownership.

The timetable for polishing up the preliminary results and running robustness tests, as well as pursuing these other lines of inquiry, would be from 6 months to a year. If additional data is required to obtain more precise measures of investor attention, then a year to 18 months might be more reasonable.

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Academic Appointments

Northwestern University, Kellogg School of Management
Visiting Professor of Finance, July 2006 – December 2006

Princeton University, Department of Economics and Bendheim Center for Finance
Assistant Professor of Economics, 2000 – 2006

Education

Ph.D., Finance, Duke University, 2001

M.A., Physics, Columbia University, 1995

B.S., Physics, University of Science and Technology of China, Hefei, China, 1993

Publications

1. “Convergence Trading with Wealth Effects: An Amplification Mechanism in Financial Markets”
Journal of Financial Economics, 2001, Vol. 62, pp. 247-292.
2. “Contagion as a Wealth Effect” (with Albert Kyle)
Journal of Finance, 2001, Vol. 56, pp. 1401-1440.
3. “Overconfidence and Speculative Bubbles” (with Jose Scheinkman)
Journal of Political Economy, 2003, Vol. 111, pp. 1183-1219.
4. “Heterogeneous Beliefs, Speculation and Trading in Financial Markets”
(with Jose Scheinkman)
Paris-Princeton Lectures on Mathematical Finance, Springer, 2003, pp. 217-250.
5. “Pay for Short-Term Performance: Executive Compensation in Speculative Markets”
(with Patrick Bolton and Jose Scheinkman)
Journal of Corporation Law, 2005, Vol. 30, pp. 721-747.
6. “Asset Float and Speculative Bubbles” (with Harrison Hong and Jose Scheinkman)
Journal of Finance, 2006, Vol. 61, pp. 1073-1117.
7. “Investor Attention, Overconfidence and Category Learning” (with Lin Peng)
Journal of Financial Economics, 2006, Vol. 80, pp. 563-602.
8. “Executive Compensation and Short-termist Behavior in Speculative Markets”
(with Patrick Bolton and Jose Scheinkman)
Review of Economic Studies, 2006, Vol. 73, pp. 577-610.
9. “Prospect Theory and Liquidation Decisions” (with Albert Kyle and Hui Ou-Yang)
Journal of Economic Theory, 2006, Vol. 129, pp. 273-288.
10. “A General Framework for Evaluating Executive Stock Options” (with Ronnie Sircar)
Journal of Economic Dynamics and Control, forthcoming.

Working Papers

- “Speculative Trading and Stock Prices: Evidence from Chinese A-B Share Premia” (with Jianping Mei and Jose Scheinkman)
Current version: December 2005
- “Time to Digest and Volatility Dynamics” (with Lin Peng)
Current version: July 2003
- “Advisors and Asset Prices: A Model of the Origins of Bubbles” (with Harrison Hong and Jose Scheinkman)
Current version: December 2005
- “What Drives the Disposition and Momentum Effects? An Analysis of a Recent Preference-Based Explanation” (with Nicholas Barberis)
Current version: June 2006
- “A Tale of Two Anomalies: The Implication of Investor Attention for Price and Earnings Momentum” (with Kewei Hou and Lin Peng)
Current version: March 2006
- “Heterogeneous Beliefs and Bond Yields” (with Hongjun Yan)
Current version: May 2006

Academic Honors

- Review of Economic Studies European Tour, May 2000
- Roger Murray Prize in 2001 Q-group meetings
- The Standard Life Investments Finance Prize for the best paper in the ECGI Finance Working Paper Series, March 2006

Professional Membership and Editorial Board

- National Bureau of Economic Research, Faculty Research Fellow, 2005 – present
- *Management Science*, Associate Editor, 2006 – present

Academic Experiences

Conference Presentations and Invited University Seminars

- 2006: American Finance Association Meetings in Boston, NBER Behavioral Finance Conference (2 papers), University of California-Davis, Federal Reserve Bank of New York, University of Chicago (GSB), Financial Intermediation Research Society Meetings in Shanghai, Bank of Italy, 12th Mitsui Life Symposium at University of Michigan, Western Finance Association Meetings in Keystone
scheduled: NBER Summer Institute Capital Markets and the Economy Conference, SITE Conference on Behavioral Economics, New York University
- 2005: American Finance Association Meetings in Philadelphia, NBER Summer Institute Asset Pricing Meeting, Cambridge/Princeton Conference on Finance, Duke/UNC Asset Pricing Conference, University of Notre Dame, Tsinghua University, Xiamen University, University of Wisconsin, World Bank
- 2004: Econometric Society Winter Meetings in San Diego, Western Finance Association Meetings in Vancouver, NBER Conference on Chinese Economy, Wharton School, Ohio State University, Cornell University, University of Kentucky, University of Florida, Rutgers University

- 2003: American Finance Association Meetings in Washington DC, Utah Winter Finance Conference, Western Finance Association Meetings in Los Cabos Mexico, Blaise Pascal Conference on Financial Modeling in Paris, European Summer Symposium in Financial Markets at Gerzensee Switzerland, NBER Corporate Finance Meeting, NBER Asset Pricing Meeting, HKUST Symposium in Finance, New York University, Columbia University, Harvard University, University of Southern California, California Institute of Technology, Carnegie Mellon University, Massachusetts Institute of Technology, University of Rochester
- 2002: Review of Financial Studies Conference at Kellogg School of Management, 2002 International Finance Conference at Tsinghua University, NBER Behavioral Finance Meeting, 5th Maryland Finance Symposium, Five Star Conference at New York University, University of Chicago, Northwestern University, Stanford University, Baruch College
- 2001: American Finance Association Meetings in New Orleans, Q-Group Spring Meeting in Tampa, University of California – Los Angeles, Rutgers University
- 2000: NASDAQ-Notre Dame Conference in Market Microstructure at University of Notre Dame, 11th Annual Conference in Financial Economics and Accounting at University of Michigan Business School, University of Chicago, Harvard University, Princeton University, Columbia University, Northwestern University, Ohio State University, Washington University (St. Louis), University of Utah, University College London, University of Toulouse, Tel Aviv University, Tilburg University

Conference Discussions

- 2006: China Industry Conference at Yale University, Financial Intermediation Research Society Meetings in Shanghai
- 2005: American Finance Association Meetings in Philadelphia, Western Finance Association Meetings in Portland (2 papers), China International Conference in Finance, NBER Summer Institute Capital Markets and the Economy Conference, Review of Financial Studies-Indiana University Conference
- 2004: American Finance Association Meetings in San Diego, Econometric Society Winter Meetings in San Diego (session chair)
- 2003: American Finance Association Meetings in Washington DC, Econometric Society Meetings in Washington DC, HKUST Conference in Corporate Finance
- 2002: American Finance Association Meetings in Atlanta
- 2001: Western Finance Association Meetings in Tucson, 7th International Finance Conference in Georgia Tech

Referee

American Economic Review, Econometrica, Economica, Journal of Economic Dynamics and Control, Journal of Economic Theory, Journal of Empirical Finance, Journal of European Economic Association, Journal of Finance, Journal of Financial Markets, Journal of Financial and Quantitative Analysis, Journal of Political Economy, Mathematical Finance, Quarterly Journal of Economics, Review of Economic Studies, Review of Financial Studies

Conference Committees

- 2006 Western Finance Meetings, 2007 American Finance Association Meetings

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Positions Held

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Assistant Professor of Finance, 2001-Present.

Educations

University of Chicago, Graduate School of Business

Ph.D., Finance, 2002.

University of Science and Technology of China (USTC)

B.S., Electrical Engineering, 1995.

Honors and Awards

Research Fellow, Charles A. Dice Center for Research in Financial Economics, Ohio State University, 2001-present

Fisher College of Business Small Grants, Ohio State University, 2006

Dean's Summer Research Fellowship, Ohio State University, 2004-2006

Dean's Teaching Performance Recognition, Ohio State University, 2003-2005

Institute for Quantitative Research in Finance (Q-Group) Research Award Grant, 2003

Grossman Fellow, University of Chicago, 2000-2001

Oscar Mayer Fellowship, University of Chicago, 2000

Graduate School of Business Fellowship, University of Chicago, 1996-2001

First Prize, Chicago Quantitative Alliance Annual Academic Paper Competition, 1999

Center for Research in Securities Prices (CRSP) Award, 1997

Publications

“Industry Information Diffusion and the Lead-Lag Effect in Stock Returns,”
forthcoming *Review of Financial Studies*

“Industry Concentration and Average Stock Returns,” 2006,
Journal of Finance 61, 1927-1956
(with David Robinson)

“Market Frictions, Price Delay, and the Cross-Section of Expected Returns,” 2005,
Review of Financial Studies 18, 981-1020, Winner of 2003 Q-Group Research Grant Award
(with Tobias Moskowitz)

“Do Investors Overvalue Firms with Bloated Balance Sheets?” 2004,
Journal of Accounting and Economics 38, 297-331
(with David Hirshleifer, Siew Hong Teoh and Yinglei Zhang)

Working Papers

“Risk Premia in Linear Factor Models: Theoretical and Econometric Issues,”
(with Robert Kimmel)

“A Tale of Two Anomalies: The Implication of Investor Attention for Price and Earnings Momentum,”
(with Lin Peng and Wei Xiong)

“What Factors Drive Global Stock Returns?”
(with Andrew Karolyi and Bong-Chan Kho)

“R² and Momentum,”
(with Lin Peng and Wei Xiong)

“The Accrual Anomaly: Risk or Mispricing?”
(with David Hirshleifer and Siew Hong Teoh)

“Accruals and Aggregate Stock Market Returns,”
(with David Hirshleifer and Siew Hong Teoh)

“Do Takeovers Increase Stockholder Value?”
(with Per Olsson and David Robinson)

“Towards a Property Rights View of Government Ownership,”
(with David Robinson)

Work in Progress

“Size Strikes Back: A Reassessment of the Size Effect”
(with Mathijs A. van Dijk)

“Characteristics versus Covariances: A Global Perspective”

(with Andrew Karolyi and Bong-Chan Kho)

“Profitability, Market Structure, and Predictable Variation in Stock Returns,”

(with David Robinson)

“Category Trading and Mergers,”

(with Lin Peng and Wei Xiong)

“Do Mergers Destroy Value? Evidence from the 1990s”

(with Per Olsson and David Robinson)

Research Interests

Asset pricing, market efficiency, behavioral finance, empirical corporate finance, capital markets research in accounting

Conference and Seminar Presentations

Presentation, American Finance Association Annual Meetings, January 2007, Chicago, Illinois

Presentation, Duke University Finance Seminar, October 2006, Durham, North Carolina

Presentation, University of North Carolina Finance Seminar, October 2006, Chapel Hill, North Carolina

Presentation, Baruch College CUNY Finance Seminar, September 2006, New York, New York

Presentation, American Accounting Association Annual Meetings, August 2006, Washington, D.C.

Presentation, Ohio State University Finance Seminar, May 2006, Columbus, Ohio

Presentation, Western Finance Association Annual Meetings, June 2005, Portland, Oregon

Discussion, American Finance Association Annual Meetings, January 2005, Philadelphia, Pennsylvania

Presentation, The First Annual Financial Research Association Conference, December 2004, Las Vegas, Nevada

Presentation, The Fifteenth Annual Conference on Financial Economics and Accounting, November 2004, Los Angeles, California

Presentation, George Mason University Finance Seminar, November 2004, Fairfax, Virginia

Presentation, The Institute for Quantitative Research in Finance (The Q-Group) Fall Seminar, October 2004, La Quinta, California

Presentation, University of Arizona Finance Seminar, October 2004, Tucson, Arizona

Session Chair, Notre Dame Behavioral Finance Conference, October 2004, Notre Dame, Indiana

Presentation, European Finance Association Annual Meetings, August 2004, Maastricht, The Netherlands

Discussion, Western Finance Association Annual Meetings, June 2004, Vancouver, Canada

Presentation, Ohio State University Finance Seminar, May 2003, Columbus, Ohio

Presentation, University of California, Santa Barbara Economics Seminar, October 2002, Santa Barbara, California

Presentation, Ohio State University Finance Seminar, February 2001, Columbus, Ohio

Presentation, Cornell University Finance Seminar, February 2001, Ithaca, New York

Presentation, University of Southern California Finance Seminar, February 2001, Los Angeles, California

Presentation, Vanderbilt University Finance Seminar, February 2001, Nashville, Tennessee

Presentation, Lehman Brothers Incorporated, January 2001, New York, New York

Presentation, University of North Carolina Finance Seminar, January 2001, Chapel Hill, North Carolina

Presentation, Yale University Finance Seminar, January 2001, New Haven, Connecticut

Presentation, Massachusetts Institute of Technology Finance Seminar, January 2001, Cambridge, Massachusetts

Presentation, Emory University Finance Seminar, January 2001, Atlanta, Georgia

Presentation, University of Texas Finance Seminar, January 2001, Austin, Texas

Presentation, American Finance Association Annual Meetings, January 2001, New Orleans, Louisiana

Presentation, University of Chicago Finance Seminar, November 2000, Chicago, Illinois

Discussion, The Sixth Annual Georgia Tech. International Finance Conference, April 2000, Atlanta, Georgia

Presentation, The Fifth Annual Georgia Tech. International Finance Conference, May 1999, Atlanta, Georgia

Presentation, University of Chicago Corporate Finance Workshop, April 1999, Chicago, Illinois

Teaching Experience

Ohio State University, Fisher College of Business, 2006
Derivatives Markets

Ohio State University, Fisher College of Business, 2002-2006
Options and Futures

University of Chicago, Graduate School of Business, 1997-2001
Investments (Teaching Assistant)

University of Chicago, Graduate School of Business, 1999-2000
Financial Instruments (Teaching Assistant)

University of Chicago, Graduate School of Business, 1997-2000
Corporate Finance (Teaching Assistant)

University of Chicago, Graduate School of Business, 1998
Theory of Financial Decisions (Teaching Assistant)

Teaching Interests

Investments, corporate finance, derivatives, asset pricing.

Ph.D. Dissertation Committee

Kuan-Hui Lee (2006, Rutgers University), Danling Jiang (2006, Florida State University), Yinglei Zhang (2005, Chinese University of Hong Kong), Christof Stahel (2004, George Mason University)

Professional Services

Referee, the *Journal of Finance*, *Review of Financial Studies*, *Journal of Empirical Finance*, *American Economic Review*, *Journal of Law, Economics, Organization*

Coordinator, Ohio State University Finance Seminar, 2006

Member of the Program Committee, Western Finance Association Annual Meetings, 2003-2006

Member of the Awards Committee, Financial Management Association Annual Meetings, 2004

Session Chair, Notre Dame Behavioral Finance Conference, October 2004, Notre Dame, Indiana

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Education

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Ph.D. in Finance, May 2002

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M.A. in Biology, May 1998

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Assistant Professor of Finance, Zicklin School of Business, Baruch College / CUNY

Research Interests

Investor Attention, Learning and Asset Pricing; Market Microstructure; Corporate Governance and Executive Compensation

Publication

- “Investor Attention, Overconfidence and Category Learning,” (with Wei Xiong), *Journal of Financial Economics*, 2006, 80, 563-602.
- “Learning with Information Capacity Constraints,” *Journal of Financial and Quantitative Analysis*, June 2005
- “Is There a Risk-Return Tradeoff? Evidence from High-Frequency Data” (with Turan Bali). *Journal of Applied Econometrics*, forthcoming.
- “Information Flow and Asset Price Comovement,” (with Tim Bollerslev and Wei Xiong). *European Financial Management*, forthcoming.
- Three chapters (“Market Makers,” “Market Liquidity,” and “The Structure of Security Markets”), with Robert A. Schwartz, in *The Encyclopedia of Finance*, C.F. Lee, Editor, Kluwer Academic Press, forthcoming.

Working Papers

- “Executive Pay, Earnings Manipulation and Shareholder Lawsuits” (with Ailsa Roell, under 2nd round review at *Review of Finance*).
- “A Tale of Two Anomalies: The Implication of Investor Attention for Price and Earnings Momentum” (with Kewei Hou and Wei Xiong, accepted by the 2007 AFA meeting).

- “The Impact of Joint Participation on Liquidity in Equity and Syndicated Bank Loan Markets” (with Linda Allen and Aron Gottesman, submitted to *Review of Financial Studies*).
- “R² and Momentum,” (with Kewei Hou and Wei Xiong)
- “Time to Digest and Volatility Dynamics” (with Wei Xiong).
- “Effects of Brokers’ Coverage Initiations on Stock Prices: Information and Beyond”.

Research in Progress

- “Market Structure and Intraday Price Volatility: An Event Study on Nasdaq’s Crosses,” with Michael Pagano and Robert Schwartz).
- “A Theory Model of Price Manipulation and Stock Market Reactions” (with Ailsa Roell).
- “Common Market Makers and Loan and Equity Market Contagion,” (with Linda Allen and Aron Gottesman)

Teaching Experience

Zicklin School of Business, Baruch College

- Fin 3710, Investment Analysis (undergraduate). Fall 2002 to Spring 2006.
- Fin 9783, Investment Analysis (Honors MBA). Fall 2005 to Spring 2006.

Professional Experience

Visiting Economist: Federal Home Loan Bank of Des Moines, Des Moines, Iowa, Summer, 1999.

Honors and Awards

- Baruch College Faculty Recognition Award for Research Excellence, 2006
- Eugene M. Lang Junior Faculty Research Fellowship, 2005-2006
- Baruch College, Zicklin School of Business, “Dean’s Teaching Excellence List” for teaching evaluation of 4.5 or higher (out of 5). 2003-2005
- Professional Staff Congress-CUNY Research Award, 2006-2007
- Professional Staff Congress-CUNY Research Award, 2005-2006
- Professional Staff Congress-CUNY Research Award, 2004-2005
- Professional Staff Congress-CUNY Research Award, 2003-2004
- Duke University Doctoral Fellowship, 1997-2002
- Ph.D. student fellowship, Wesleyan University, 1995-1997

Conference Presentations

- American Finance Association Meetings, Jan. 2007 (scheduled). “A Tale of Two Momentum Anomalies: the Effect of Investor Attention on Stock Market Under- and Overreactions”.

- Conference on Banking, Corporate Finance and Intermediation, June, 2006, Shanghai, China. “Executive Pay and Shareholder Lawsuits”.
- Valuation in Financial Markets, University of California, Davis, April, 2006. “A Tale of Two Anomalies: The Implication of Investor Attention for Price and Earnings Momentum”.*
- Western Finance Association Meetings, Portland, Oregon, June 2005. “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- American Finance Association Meetings, Philadelphia, PA. Jan. 2005. “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- Center for Economic Policy Research (CEPR) European Summer Symposium in Financial Markets, Gerzensee, Switzerland July 2004. “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- The Financial Intermediation Society Conference on Banking, Insurance and Intermediation, Capri, Italy, May 2004. “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.*
- Corporate Governance at the Crossroads, University of Connecticut School of Law, April 2004. “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- Western Finance Association Meetings, Los Cabos, Mexico June 2003. “Capacity constrained learning and asset price comovements”.*
- Summer Meeting of Econometric Society, Chicago, IL, June 2003. “Learning with Information Capacity Constraints”.
- Summer Meeting of Econometric Society, Chicago, IL, June 2003. “Time to Digest and Volatility Dynamics”.
- The Microstructure Conference in Honor of David K. Whitcomb, Newark, NJ, Oct. 2002. “Capacity Constrained Learning and Asset Price Comovement”.
- Frank Batten Young Scholars Conference, Williamsburg, June, 2002. “Learning with Information Capacity Constraints”.
- Frank Batten Young Scholars Conference, Williamsburg, June, 2002. “Time to Digest and Volatility Dynamics”.
- Washington Area Finance Association Meetings, Washington DC, April 2002. “Capacity Constrained Learning and Asset Price Comovement”.
- Midwest Finance Association Meetings, Chicago, IL, March 2002. “Time to Digest and Volatility Dynamics”.
- Conference on Financial Economics and Accounting, New Brunswick, NJ, Sep. 2001. “Time to Digest and Volatility Dynamics”.

* Presented by coauthor.

Seminar Presentations

- Baruch College, March 2006, “A Tale of Two Anomalies: The Implication of Investor Attention for Price and Earnings Momentum”.

- Baruch College, Dec. 2005, “Information Diffusion and Volatility Dynamics”.
- Peking University, June 2005, “Executive Pay and Shareholder Lawsuits”.
- University of Notre Dame, April 2005, “ R^2 and Momentum”.
- Baruch College, March 2005, “ R^2 and Momentum”.
- University of Washington, Seattle, Nov. 2004, “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- Northwestern University, Aug. 2004, “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- Ohio State University, June 2004, “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- Baruch College, March 2004, “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- University of Hong Kong, January 2004, “Executive Pay, Earnings Manipulation and Shareholder Lawsuits”.
- Baruch College, November 2003, “An Investigation of Risk-Return Tradeoff”.
- Princeton University, July 2003, “Executive Pay and Shareholder Lawsuits”.
- Baruch College, April 2003, “Information Flow and Asset Price Comovement”.
- UNC-Chapel Hill macroeconomics workshop, March 2002, “Time to Digest and Volatility Dynamics”.
- Baruch College, Feb. 2002, “Learning with Information Capacity Constraints”.
- Columbia University, Feb. 2002, “Learning with Information Capacity Constraints”.
- Cornell University, Feb. 2002, “Learning with Information Capacity Constraints”.
- Indiana University at Bloomington, Feb. 2002, “Learning with Information Capacity Constraints”.
- Northwestern University, Feb. 2002, “Learning with Information Capacity Constraints”.
- NYU, Feb. 2002, “Learning with Information Capacity Constraints”.
- UNC-CH Kenan-Flagler Business School, Feb. 2002, “Learning with Information Capacity Constraints”.
- Yale University, Feb. 2002, “Learning with Information Capacity Constraints”.
- Federal Reserve Bank, Board of Governors, Jan. 2002, “Learning with Information Capacity Constraints”.
- Penn. State University, Jan. 2002, “Learning with Information Capacity Constraints”.
- University of Notre Dame, Jan. 2002, “Learning with Information Capacity Constraints”.
- Duke University econometrics workshop, Nov. 2001, “Learning with Information Capacity Constraints”.
- Duke University Finance workshop, Oct. 2001, “Learning with Information Capacity Constraints”.

- Duke University Finance workshop, June 2001, “Time to Digest and Volatility Dynamics”.
- Duke University econometrics workshop, May 2001, “Time to Digest and Volatility Dynamics”.

Conference Discussions

- North American Winter Meeting of the Econometric Society, Boston, MA, Jan. 2006. “Information Acquisition and Portfolio Under-diversification,” Stijn Van Nieuwerburgh and Laura Veldkamp.
- International Finance Conference, Kunming, China. July 2005. “Search and Endogenous Concentration of Liquidity in Asset Markets,” Dimitri Vayanos and Tan Wang.
- American Finance Association Meetings, Washington, DC, Jan. 2003. “News Spillovers in the Sovereign Debt Market,” Amar Gande and David Parsley.
- International Finance Conference, Beijing, China, July, 2002. “Empirical Analysis of Predicting Financial Distress for Publicly Traded Companies,” Houqi Zhang, Yuepin Liu and Mingbo Jiang.
- Midwest Finance Association Meetings, Chicago, IL, March 2002. “Value at Risk Equilibrium Pricing Model,” Guy Kaplanski and Kvoll Yovam.

Professional Activities

- **Referee** for *Journal of Political Economy*, *Journal of Finance*, *Review of Financial Studies*, *Journal of the European Economic Association*, *Review of Finance*, *Journal of Financial Markets*, *Journal of Empirical Finance*, *Journal of Financial Econometrics*, *Southern Economic Journal*.
- **Session Chair/Moderator:**
 - The 2003 Summer Meeting of Econometric Society, Chicago, IL, June 2003. Session Chair: “Uncertainty and Information: Theory”
 - Baruch College Conference: Coping with Institutional Order Flow, NY, NY, April 2003. Session Moderator: “Coping with Institutional Trading Practices”
 - Baruch College Equity Markets Seminar, NY, NY, November 2003. Session Moderator: “Order Driven & Quote Driven Markets” (with Robert A. Schwartz)
- **Member** of Eugene Lang Fellowship Committee (2006-2007)
- **Member** of American Finance Association, Western Finance Association, Econometric Society